§ Kostant slices.

Prop. $\mathcal{C} \cong f + \mathcal{Z}_{\mathfrak{g}}(e) \subset f + \mathfrak{b}$.

Proof.

$\forall x \in f + \mathcal{Z}_{\mathfrak{g}}(e) \subset f + \mathfrak{b}$ by Kostant's theorem exists a unique pair $(g, a) \in N \times \mathcal{C}$: $g.a.g^{-1} = x$.
This gives a map

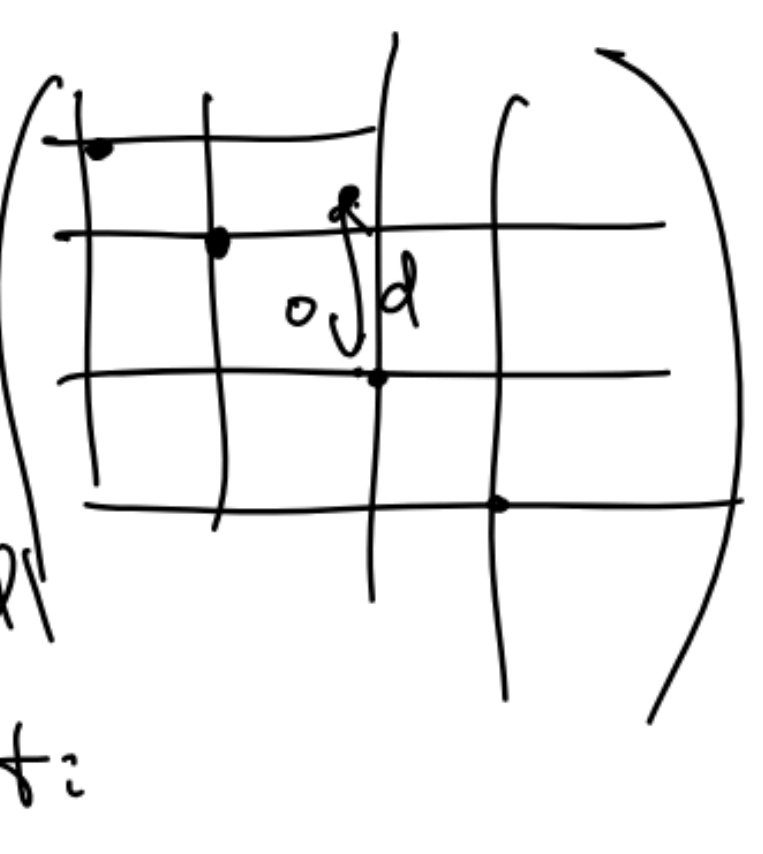
$$\begin{tikzcd} f + \mathcal{Z}_{\mathfrak{g}}(e) \arrow[r, hook, "\iota"] \arrow[dr, "\Psi"'] & f + \mathfrak{b} \cong N \times \mathcal{C} \arrow[d, "\pi_2"] \\ & \mathcal{C} \end{tikzcd}$$

$$x \in f + \mathcal{Z}_{\mathfrak{g}}(e) \Rightarrow x \sim \begin{pmatrix} b_0 & b_1 & \cdots & b_{n-1} \\ (n-1) & b_0 & \ddots & \vdots \\ & 2(n-2) & \ddots & b_1 \\ 0 & & (n-1)\cdot 1 & b_0 \end{pmatrix}.$$

We study $\det(\lambda - x)$.

Claim: The coeff of λ^{n-k} has the form (Here $k \geq 1$)
$\underset{\neq 0}{c_k} b_{k-1} + p_k(b_1, \dots, b_{k-2})$.

Proof.
To get λ^{n-k} at least $n-k$ terms should be taken from the diagonal. We have $\sum_{i=1}^{n} \sigma(i) = \frac{n(n+1)}{2}$.



Hence if b_m appears $\Rightarrow$ for at least d indices $i : \sigma(i) = i+1$.
Hence maximal possible m is k and for such term we have σ such that:
1) σ has $n-k$ fixed points.
2) $\exists\, i_1 < \dots < i_{k-1} : \sigma(i_j) = i_j + 1$.

This implies that σ has the form
$(i\ i{+}1 \dots i{+}k{+}1)$.
All such go with coeff. of the same sign by b_{k-1}.
$\blacksquare$

Therefore the map Ψ is described by
$a_{n-k} = \underset{\neq 0}{c_k} b_{k-1} + p_k(b_0, \dots, b_{k-2})$. $\Rightarrow$ is an isomorphism. $\blacksquare$

Def. $f + \mathcal{Z}_{\mathfrak{g}}(e)$ is called Kostant slice.

Remarks.

- Kostant slice is also transverse to orbits.

§ Slodowy slices.

Def. Let $e \in \mathfrak{g}$ be a nilpotent element.
Let (e, h, f) be a corresp $\mathfrak{sl}_2$-triple.
$S_f := f + Z_{\mathfrak{g}}(e)$.

Proposition.

(i) S_f is transverse to orbits $\mathcal{O}_x$, namely, if $x \in S_f$, $T_x \mathcal{O}_x + T_x S_f = T_x \mathfrak{g} \cong \mathfrak{g}$.

(ii) $S_f \cap \mathcal{O}_f = \{f\}$.

Proof.

(i) First for $x = f$.
$T_f \mathcal{O}_f = \mathrm{Im}(\mathrm{ad}_f)$, $T_f S_f = Z_{\mathfrak{g}}(e) = \mathrm{Ker}(\mathrm{ad}_e)$.
For any f.d. $\mathfrak{sl}_2$-rep: $W = \mathrm{Im}(\rho(f)) \oplus \mathrm{Ker}(\rho(e))$.
In general, $x = f + s$, $s \in \mathrm{Ker}(\mathrm{ad}(e))$.
$T_x \mathcal{O}_x = \mathrm{Im}(\mathrm{ad}_x)$, $T_x(S_f) = \mathrm{Ker}(\mathrm{ad}_e)$ are transverse in some nbhd of $x \in \mathfrak{g}$.

$\mathfrak{sl}_2$-triple induces a homomorphism $\gamma : SL_2(\mathbb{C}) \longrightarrow G$.
We have adjoint action of $h_t = \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix}$
$\mathrm{Ad}_{\gamma(h_t)}(f) = t^{-2} \cdot f$.
Define $\mathbb{C}^\times$ action on S_f by
$t : f + s \longmapsto t^{2} \cdot \mathrm{Ad}_{\gamma(t^{-1})}(f+s) = f + t^{-2} \mathrm{Ad}_{\gamma(t^{-1})}(s)$.
$s \in \mathrm{Ker}(\mathrm{ad}(f)) \Rightarrow h_t$ weights are $\leq 0 \Rightarrow t$-weights of s are $> 0 \Rightarrow \lim_{t \to 0} (t.(f+s)) = f \ \forall f + s \in S_e$.
Therefore since the action preserves transversality, in a nbhd intersection is transverse and action contracts S_f to f $\Rightarrow$ $\blacksquare$

(ii)

$\boxed{\text{Fact: } \mathcal{O}_f \text{ is } \mathbb{C}^\times \text{ stable}}$

Note that $\mathcal{O}_f$ is $\mathbb{C}^\times$-invariant $\Rightarrow$ $t^{-2} \cdot \mathrm{Ad}_{\gamma(t^{-1})}(\mathcal{O}_f) \subset \mathcal{O}_f$.
Moreover S_f is $\mathbb{C}^\times$-invariant.
Then $\mathcal{O}_f \cap S_f$ is $\mathbb{C}^\times$ invariant.
Then if $x \in \mathcal{O}_f \cap S_f \setminus \{f\}$, $\lim_{t \to 0}(t.x) = f \Rightarrow$
$\mathcal{O}_f \cap S_f$ intersects arbitrarily small nbhds of f / there is a curve on $\mathcal{O}_f \cap S_f$ through f.
But $\mathcal{O}_f$ and S_f are strictly transverse at f. $\blacksquare$

Remark: Away from f intersection in general is not strictly transverse.
Dimension of $\mathcal{O}_G(f)$ is complementary to $\dim(S_f)$.

But this is not the case for $\mathcal{O}_G(x)$, $x \in S_f$ in general.

Example $G = SL_2$, $\mathfrak{g} = \mathfrak{sl}_2$

$S_f = \{ f + x \cdot e \mid x \in \mathbb{C} \}$.

Generic orbits are defined by:
$\mathcal{O}_\Delta = \left\{ \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \,\middle|\, a^2 + bc = \Delta \right\}$

Intersections: $c = 1$, $a = 0 \Rightarrow x = b = \Delta$ - unique int.
On $\mathcal{O} = \{0\} \Rightarrow$ no intersections.

$T_{\begin{pmatrix} 0 & x \\ 1 & 0 \end{pmatrix}}(\mathcal{O}) = \mathrm{Ker}(db + x\,dc)$

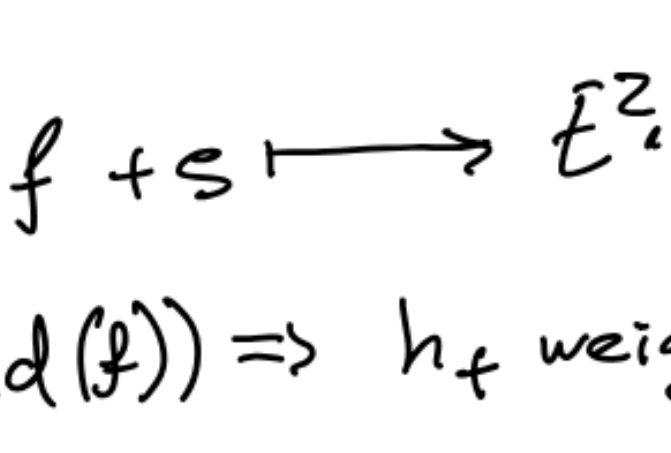
$T_{\begin{pmatrix} 0 & x \\ 1 & 0 \end{pmatrix}}(S_f) = \langle e \rangle = \langle \frac{\partial}{\partial b} \rangle$.

$\langle db + x\,dc, \frac{\partial}{\partial b} \rangle \equiv 1 \Rightarrow$ Always transverse.

Example: $G = SL_3(\mathbb{C})$, $\mathfrak{g} = \mathfrak{sl}_3(\mathbb{C})$, subregular orbit.

$f = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $e = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $\mathcal{Z}_{\mathfrak{g}}(e) = \langle e, \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \rangle$

$\mathfrak{sl}_2$ decomp:



Hence $S_f = \left\{ X = \begin{pmatrix} -x_0 & x_3 & x_1 \\ 1 & -x_0 & 0 \\ 0 & x_2 & 2x_0 \end{pmatrix} \,\middle|\, x_i \in \mathbb{C} \right\}$.

$\mathrm{Tr}(X^2) = 6x_0^2 + x_3 + x_1 x_2$; $\det(X) = 2x_0^3 + x_1 x_2 - 2x_0 x_3$.

§ Poisson structure on S_f via reduction.

- Decompose $\mathfrak{g} = \bigoplus_{i \in \mathbb{Z}} \mathfrak{g}_i$ - h-weight decomposition.
- Define $\omega_S : \Lambda^2 \mathfrak{g}_1 \longrightarrow \mathbb{C}$
  $x \wedge y \longmapsto \chi([x,y]) = \langle f, [x,y] \rangle$

Prop. ω_S is a symplectic 2-form on $\mathfrak{g}_1$.

Proof. Non-degeneracy:
$x \in \mathfrak{g}_1^{\angle} \Leftrightarrow [f, x] \in \mathfrak{g}_1^{\perp}$.
But $\mathrm{ad}_f : \mathfrak{g}_1 \xrightarrow{\sim} \mathfrak{g}_{-1}$ and $\langle \cdot, \cdot \rangle|_{\mathfrak{g}_1 \otimes \mathfrak{g}_{-1}}$ is non-deg. $\Rightarrow x = 0$
$\blacksquare$.

- Terminology/Notation:

Let V be a vector space, ω a symplectic form on V, then for a subspace $W \subset V$,

- $W^{\angle} := \{ u \in V \mid \omega(u, w) = 0 \ \forall w \in W \}$.

A subspace $W \subset V$ is

- Isotropic if $W \subseteq W^{\angle}$.
- Coisotropic if $W^{\angle} \subseteq W$.
- Lagrangian if $W^{\angle} = W$ (isotropic and coisotropic).

Fix ℓ - any lagrangian subspace of $\mathfrak{g}_1$.

Remark If f-principal nilpotent, $\mathfrak{g}_1 = \{0\}$.

Set $\mathfrak{m} = \ell \oplus \bigoplus_{i \geq 2} \mathfrak{g}_i$. Set $M \subset G$: $\mathrm{Lie}(M) = \mathfrak{m}$, M is connected.

- $\mathfrak{m}$ is a nilpotent Lie algebra.

DrawNote